Fixed Heavy-Tailed Probabilistic Forecasting under Missing Historical Observations

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Abstract

Probabilistic forecasting from missing historical observations requires both accurate predictions and uncertainty estimates that remain meaningful under controlled input corruption. We introduce a leakage-controlled fixed missing-history evaluation protocol with train-only normalization, fixed validation and test masks, and no masking of future targets. Within this protocol, we study a lightweight fixed Student-t head with degrees of freedom fixed at 8. It has the same trainable parameter count as matched Gaussian and Laplace heads. On TimePerceiver, Student-t variants improve negative log-likelihood (NLL) and continuous ranked probability score (CRPS) across three datasets and four probability-head training seeds, with all 12 paired comparisons meeting the predefined point-noninferiority criterion. A distribution-head ablation shows that fixed df=8 is sufficient under the evaluated setting, while learned per-channel degrees of freedom add no consistent benefit. On PatchTST, the fixed Student-t head improves NLL and CRPS on all three datasets. However, Electricity mean squared error (MSE) increases by 1.194%, slightly exceeding the pre-registered 1% tolerance, so the complete second-backbone gate is not positive. We also retain two negative findings: global-statistics feature-wise linear modulation does not outperform a capacity-matched static adapter, and additive split conformal calibration does not improve native intervals in the tested condition. The evidence is limited to two backbones, three datasets, fixed masks, and restricted seed coverage; it supports probabilistic-score robustness rather than complete metric-wise invariance.

1 Introduction

Forecasting systems often receive incomplete histories. Sensors fail, communication channels drop observations, and data pipelines can produce blocks or channels of missing values. A useful forecasting model must therefore operate on a partially observed historical window and quantify uncertainty about an unobserved future. These requirements are distinct: a point forecast can remain plausible while its predictive distribution becomes too narrow, too light-tailed, or otherwise poorly matched to the residuals.

Many missing-data studies focus on reconstructing the history before evaluating a downstream task. Imputation is important, but reconstruction quality alone does not determine forecast quality. A forecasting-centered evaluation must hold the historical mask fixed across methods, keep future targets outside the input, and prevent test statistics from entering normalization or model selection. Without these controls, apparent differences can reflect mask resampling or leakage rather than the predictive method.

Gaussian likelihoods are attractive because they are simple and differentiable. Their tails can nevertheless assign excessive negative log-likelihood to large standardized residuals. A Student-t likelihood offers heavier tails while retaining a familiar location-scale parameterization. Fixing its degrees of freedom removes the need to estimate an additional shape parameter and makes

capacity matching straightforward. Proper scoring rules, including log score and the continuous ranked probability score, provide complementary views of probabilistic quality [2].

This study combines a fixed missing-history protocol with a lightweight fixed Student-t head. The historical observation mask is treated as an evaluation asset rather than an ephemeral random draw. Gaussian, Laplace, and fixed Student-t heads use matched trainable location-scale projections. We retain negative results for a missingness-aware feature-wise linear modulation (FiLM) adapter and an additive split-conformal procedure. The goal is an auditable account of what improves, what does not, and where a tradeoff remains.

The main evidence uses TimePerceiver [3] on ETTh1, Weather, and Electricity. Across four probability-head training seeds, Student-t variants improve NLL and CRPS in all 12 paired cells and meet point noninferiority in all 12. A controlled ablation supports fixed df=8 as sufficient. A second study uses PatchTST [4]. Its fixed Student-t head improves NLL and CRPS on all three datasets, but Electricity MSE rises from 0.165829 to 0.167809. The 1.194% increase exceeds the pre-registered 1% tolerance, so the full multi-metric gate remains negative.

The contributions are threefold:

1. We define a leakage-controlled fixed missing-history evaluation protocol for deterministic and probabilistic forecasting.
2. We study a lightweight fixed Student-t probability head with the same trainable parameter count as matched Gaussian and Laplace heads.
3. We provide a cross-backbone empirical study with a distribution ablation, retained negative results, and an explicit PatchTST Electricity point-prediction tradeoff.

2 Related Work

2.1 Probabilistic time-series forecasting

Neural probabilistic forecasters learn predictive distributions rather than only conditional means. DeepAR, for example, trains a shared autoregressive recurrent model over related series and optimizes a parametric likelihood [5]. Such models make distributional assumptions explicit and enable evaluation with proper scoring rules. Log score assesses predictive density at the realized target, while CRPS evaluates the full univariate cumulative distribution [2]. Our study adopts both scores and also reports point metrics from the predictive location.

2.2 Heavy-tailed predictive distributions

Location-scale likelihoods differ primarily in how they allocate mass away from the center. A Gaussian density decays rapidly, a Laplace density has exponential tails, and a Student-t density has polynomial tails. The degrees of freedom control the Student-t tail weight. We compare these families under matched location-scale capacity. Unlike an adaptive shape model, our selected form fixes df=8 and introduces no trainable degrees-of-freedom parameter.

2.3 Forecasting with missing historical observations

Time-series missingness has often been addressed through imputation. BRITS learns missing values in a bidirectional recurrent system and evaluates downstream prediction tasks [1]. CSDI instead

models a conditional distribution over missing values with score-based diffusion and supports probabilistic imputation [7]. These methods motivate careful treatment of observation masks. Our protocol asks a narrower question: given a fixed incomplete history, how do point and probabilistic forecast metrics change when the forecasting head changes? We do not present a new imputation method.

2.4 Conformal prediction for time series

Conformal methods transform calibration residuals into prediction sets under explicit assumptions about calibration and test examples. Time-series extensions address multi-horizon forecasting and weakened exchangeability conditions [6]. Our conformal experiment is deliberately limited to an additive split procedure with a strict temporal split and embargo. Its negative outcome is specific to that design and does not generalize to other conformal scores or online methods.

2.5 General-purpose forecasting backbones

TimePerceiver uses latent bottlenecks and timestamp-aligned decoder queries to support multiple temporal prediction layouts [3]. PatchTST tokenizes temporal patches and shares Transformer weights across channels [4]. They provide structurally different frozen backbones for the present head comparison. We treat their clean checkpoints as fixed feature extractors and avoid ranking the two architectures by their absolute metrics.

3 Problem Formulation

Let $X \in \mathbb{R}^{L \times C}$ denote a historical multivariate sequence with input length L and C channels. Let $Y \in \mathbb{R}^{H \times C}$ denote the future target over forecast horizon H. A binary historical observation mask $M \in \{0,1\}^{L \times C}$ indicates which elements of X are observed. The masked historical input is

$$\widetilde{X} = M \odot X + (1 - M) \odot v_{\text{miss}}, \tag{1}$$

where v_{miss} is the protocol's missing-value representation and $\odot$ denotes elementwise multiplication.

The mask acts only on historical inputs. The future target Y is never masked, copied into $\widetilde{X}$, or used to construct the input mask. Normalization parameters are fitted on the training split only and then reused for validation and test data. Validation and test masks are fixed assets, so matched methods receive identical observed histories.

A backbone maps $\widetilde{X}$, time features, and the observation mask to forecast features Z. A probability head then defines a marginal predictive density for each target element:

$$p(Y \mid \widetilde{X}, M) = \prod_{h=1}^{H} \prod_{c=1}^{C} p(Y_{h,c} \mid Z_{h,c}) \,. \tag{2}$$

Equation 2 is a modeling factorization for marginal target distributions. It does not represent cross-channel or cross-horizon covariance, and its marginal intervals do not imply simultaneous coverage.

We evaluate point accuracy with mean squared error (MSE), mean absolute error (MAE), and root mean squared error (RMSE). We evaluate predictive distributions with negative log-likelihood

(NLL), CRPS, interval coverage, interval width, and interval score. Lower values are better for all listed metrics except coverage, which is compared with its nominal target.

For a candidate head a and a matched Gaussian reference g, point noninferiority requires

$$\frac{\mathrm{MSE}_a - \mathrm{MSE}_g}{\mathrm{MSE}_g} \leq 0.01. \tag{3}$$

The 1% threshold is pre-registered for the paired comparisons and is not changed after observing test results.

4 Method

4.1 Backbone and probability-head interface

The method operates on forecast features produced by a frozen backbone. For TimePerceiver, these features follow the model's latent encoder and query-based decoder [3]. For PatchTST, they follow the supervised patch encoder [4]. We do not alter either backbone architecture. A small output head maps each feature vector to the parameters of a marginal predictive distribution.

4.2 Gaussian reference

The Gaussian reference predicts a location μ and an unconstrained scale value s. Its positive scale is

$$\sigma = \mathrm{softplus}(s) + 10^{-4}. \tag{4}$$

The head is trained by minimizing mean Gaussian NLL over observed target elements. Because future targets are complete in this protocol, the target mask does not remove forecast elements.

4.3 Fixed Student-t head

The fixed Student-t head predicts the same location μ and scale σ as the Gaussian reference. It sets the degrees of freedom to

$$\nu = 8 \tag{5}$$

for every target channel and horizon. There is no trainable degrees-of-freedom parameter. The NLL objective follows the univariate Student-t density independently for each target element.

The fixed value is a reference setting shown to be empirically sufficient by the distribution-head ablation before the PatchTST experiment. It was retained for Gate 5b without dataset-specific retuning. We do not describe df=8 as theoretically optimal.

4.4 Capacity matching

Gaussian, Laplace, and fixed Student-t heads use identical trainable location and scale projections. They therefore have the same trainable parameter count for a given backbone and dataset. Their difference lies in the likelihood and interval quantiles, not in the number of optimized weights. A learned-df Student-t variant adds one degrees-of-freedom parameter per target channel and is used only in the controlled ablation and TimePerceiver multi-seed evidence.

Table 1: Clean-input TimePerceiver references. Metrics are reproduced under the project protocol with one official training seed per dataset.

Dataset	MSE	MAE	RMSE	Best epoch
ETTh1	0.374462	0.398943	0.611933	26
Weather	0.144059	0.199637	0.379550	38
Electricity	0.125364	0.220123	0.354068	60

4.5 Evaluation protocol

Every paired head comparison uses the same backbone checkpoint, train/validation/test split, normalization statistics, validation mask, test mask, optimization budget, and checkpoint-selection rule. The test set is evaluated only after selecting the lowest validation-NLL checkpoint. NLL and CRPS provide the primary probabilistic comparison, while Equation 3 guards against a material point-prediction regression.

5 Experimental Setup

5.1 Datasets and deterministic references

We evaluate ETTh1, Weather, and Electricity. Every experiment uses input length 384 and prediction length 96. We retain the official train, validation, and test boundaries for each dataset. Normalization statistics are fitted on the training split only and then frozen. Table 1 records the clean-input TimePerceiver references used to quantify degradation under missing history.

5.2 Missing-history protocol

Historical masking uses random, block, channel, and mixed masks at missing rates 0.1, 0.3, 0.5, and 0.7. Validation and test masks are generated once and reused across matched models. The fixed test-mask seed is 2025. Training masks follow the recorded deterministic seed rule for each formal experiment. Missingness changes only the historical input. Future targets remain complete and are never available to the model.

5.3 Probability-head training

TimePerceiver comparisons keep the official backbone checkpoint fixed and vary the probability-head training seed over 2021, 2022, 2023, and 2025. This is a four-seed head study rather than full end-to-end backbone retraining. The broader cross-condition and distribution-ablation experiments use seed 2025 unless stated otherwise.

PatchTST uses one clean-backbone training seed and one probability-head seed, both 2025. Its second-backbone comparison is limited to random missingness at rate 0.3. Gaussian and fixed Student-t heads share the backbone checkpoint, split, masks, optimizer budget, and validation-based checkpoint rule.

Table 2: Forecast degradation under fixed historical masks. Relative degradation is measured against the corresponding clean TimePerceiver reference.

Dataset	Mask	Rate	MSE	Clean degradation
ETTh1	Block	0.30	0.423837	13.19%
ETTh1	Channel	0.30	0.583704	55.88%
ETTh1	Mixed	0.30	0.495055	32.20%
ETTh1	Random	0.10	0.378811	1.16%
ETTh1	Random	0.30	0.425394	13.60%
ETTh1	Random	0.50	0.525856	40.43%
Electricity	Block	0.30	0.184630	47.28%
Electricity	Channel	0.30	0.391038	211.92%
Electricity	Mixed	0.30	0.246785	96.86%
Electricity	Random	0.10	0.138429	10.42%
Electricity	Random	0.30	0.200572	59.99%
Electricity	Random	0.50	0.325732	159.83%
Weather	Block	0.30	0.180420	25.24%
Weather	Channel	0.30	0.284471	97.47%
Weather	Mixed	0.30	0.224939	56.14%
Weather	Random	0.10	0.169631	17.75%
Weather	Random	0.30	0.240810	67.16%
Weather	Random	0.50	0.331313	129.99%

5.4 Metrics and model selection

We report NLL, CRPS, MSE, MAE, RMSE, prediction-interval coverage probability (PICP), mean prediction-interval width (MPIW), and interval score. CRPS uses 64 Monte Carlo (MC) samples with MC seed 52025. Native intervals come directly from each predictive distribution. Early stopping and checkpoint selection use validation NLL. Test metrics are not used for training, hyperparameter selection, or early stopping.

For paired gates, lower NLL is a probabilistic win. CRPS must be noninferior under the registered comparison rule, and relative MSE degradation must not exceed 1%. We report each component separately rather than collapsing a partial pass into a positive overall decision.

6 Results

6.1 Deterministic degradation under missing history

The fixed-mask deterministic audit contains 48 dataset, mask-type, and missing-rate cells. Across the 12 dataset-by-mask-type curves, MSE increases monotonically over the evaluated missing rates. Table 2 gives the representative matrix retained in the main text, and Figure 1 visualizes the degradation. The complete matrix appears in Appendix A.1.

6.2 TimePerceiver probabilistic forecasting

The primary TimePerceiver matrix contains 12 paired cells: three datasets crossed with four probability-head training seeds. Student-t variants record 12/12 NLL wins, 12/12 CRPS-noninferior outcomes, 12/12 point-noninferior outcomes, and 12/12 full paired passes. All 3/3 dataset-level mean gates pass. Table 3 reports the dataset means and standard deviations.

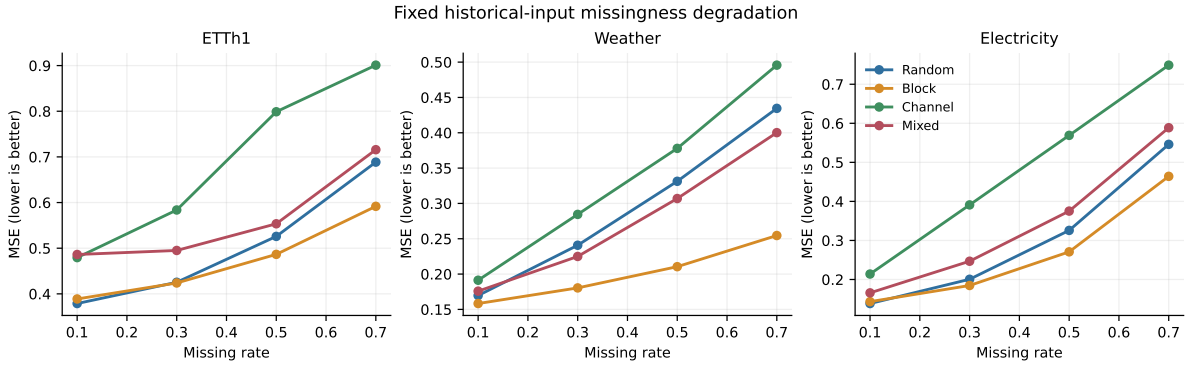


Figure 1: MSE under the fixed missing-history evaluation. Higher missing rates increase error across the evaluated datasets and mask types.

Table 3: TimePerceiver four-head-seed comparison. Deltas are Student-t minus Gaussian; negative values favor Student-t.

Dataset	G NLL	T NLL	ΔNLL	NLL wins	G CRPS	T CRPS	ΔCRPS	G MSE	T MSE	ΔMSE	Point NI
ETTh1	0.915014	0.871113	-0.043900	4	0.342011	0.337656	-0.004356	0.416862	0.415611	-0.001252	4
Weather	0.577226	0.103670	-0.473557	4	0.244788	0.207544	-0.037245	0.210306	0.209818	-0.000488	4
Electricity	0.439644	0.365348	-0.074296	4	0.209365	0.207395	-0.001970	0.167419	0.167703	0.000284	4

Figure 2 shows paired NLL changes for every dataset and seed. Figure 3 summarizes NLL, CRPS, and MSE means. The evidence concerns probability-head optimization on fixed official backbone checkpoints, so it does not measure variability from backbone retraining.

6.3 Distribution-head ablation

Table 4 compares Gaussian, Laplace, fixed Student-t, and learned-df Student-t heads under random missingness at rate 0.3. The fixed Student-t head records 3/3 NLL wins, 3/3 CRPS-noninferior outcomes, and 3/3 point-noninferior outcomes against Gaussian. The learned-df variant does not consistently improve over fixed df=8. Laplace also lacks equal support across all paired criteria. The formal conclusion is therefore that a fixed heavy-tailed likelihood is sufficient in the evaluated scope.

The fixed Student-t head has the same trainable parameter count as Gaussian and Laplace. Table 5 reports the controlled parameter, latency, and memory measurements. The learned-df head adds only per-channel shape parameters, yet this extra flexibility does not produce a consistent metric benefit.

6.4 Second-backbone replication

The PatchTST comparison records 3/3 NLL wins, 3/3 CRPS-noninferior outcomes, 2/3 point-noninferior outcomes, and 2/3 full paired passes for the fixed Student-t head. Table 6 reports the paired deltas, and Figure 4 visualizes their signs. ETTh1 and Weather pass every component.

Electricity does not pass the point criterion. Gaussian MSE is 0.165829 and fixed Student-t MSE is 0.167809, a change of +0.001980 or +1.194%. The increase is above the pre-registered 1% tolerance in Equation 3. The decision is therefore `SECOND_BACKBONE_POINT_TRADEOFF`. The NLL

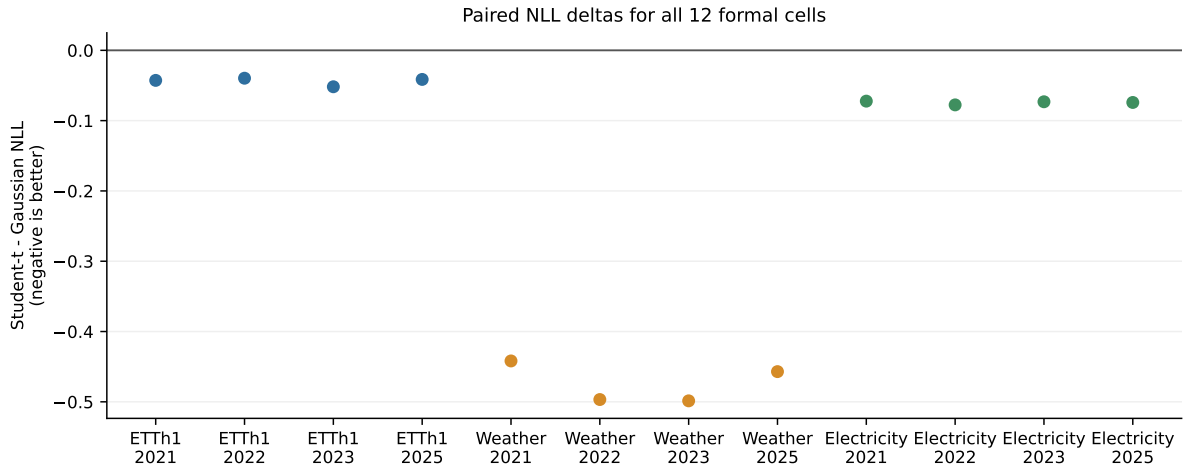


Figure 2: Paired NLL deltas for the TimePerceiver multi-seed matrix. Every displayed delta is negative.

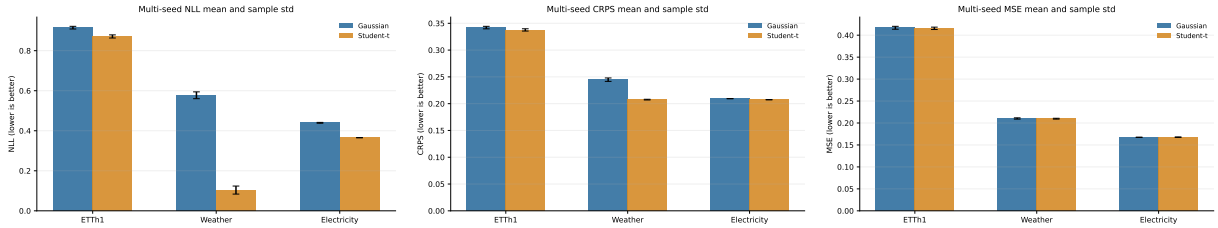


Figure 3: TimePerceiver dataset-level mean and standard-deviation summaries for NLL, CRPS, and MSE across four probability-head training seeds.

and CRPS direction transfers to PatchTST, but the complete multi-metric gate does not.

6.5 Heavy-tail diagnostics

Residual-percentile diagnostics indicate that much of the NLL gain appears in the most extreme Gaussian-residual region. Figure 5 shows this pattern for the TimePerceiver ablation. The corresponding PatchTST diagnostic covers approximately 1.82×10^8 forecast elements and yields the same qualitative concentration in its most extreme bin. These analyses are descriptive. They are not formal gate evidence, causal mechanism tests, or inferential samples of 1.82×10^8 independent observations.

7 Ablations and Negative Results

7.1 Missingness-aware FiLM

The frozen-backbone static adapter improves ETTh1 random-0.3 MSE from 0.425394 to 0.361377. A missingness-aware FiLM adapter reaches 0.363156, which is 0.001778 worse than the static adapter. The mechanism audit finds a nonconstant condition representation, nonzero FiLM modulation, finite gradients, and updated conditioning parameters. These checks do not support an

Table 4: Distribution-head ablation on frozen TimePerceiver checkpoints. The fixed Student-t head uses df=8.

Dataset	Head	NLL	CRPS	MSE	Coverage error	Trainable
etth1	Gaussian	0.918535	0.344127	0.420332	0.038132	6,168
etth1	Laplace	0.879116	0.341785	0.424149	0.011910	6,168
etth1	Fixed Student-t (df=8)	0.864949	0.336548	0.413927	0.022006	6,168
etth1	Learned Student-t	0.877154	0.340263	0.419177	0.020301	6,175
weather	Gaussian	0.586325	0.245442	0.212237	0.082667	24,624
weather	Laplace	0.204399	0.209589	0.210051	0.070806	24,624
weather	Fixed Student-t (df=8)	0.136165	0.208461	0.210994	0.051632	24,624
weather	Learned Student-t	0.129293	0.208276	0.210836	0.052156	24,645
electricity	Gaussian	0.440598	0.209453	0.167881	0.038304	24,624
electricity	Laplace	0.370388	0.208768	0.170436	0.018222	24,624
electricity	Fixed Student-t (df=8)	0.367009	0.207797	0.168473	0.018744	24,624
electricity	Learned Student-t	0.366449	0.207786	0.168486	0.018502	24,945

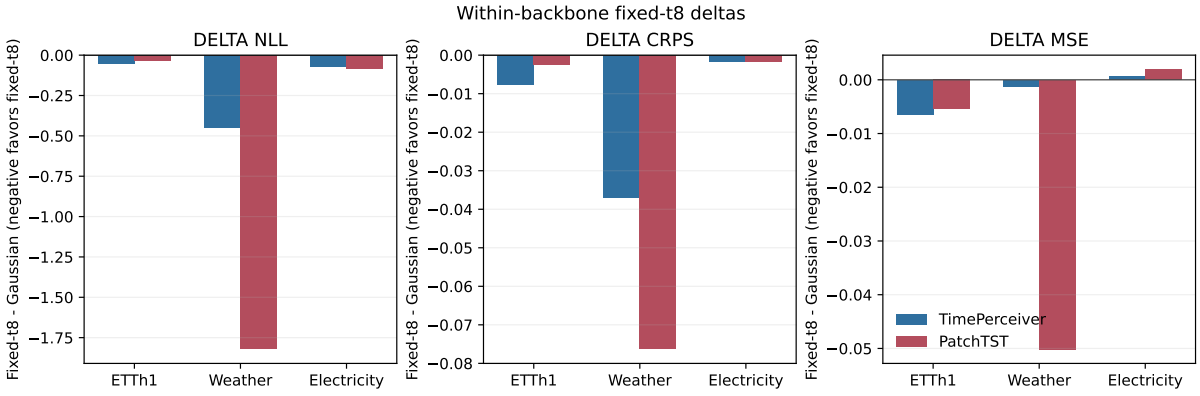


Figure 4: Within-backbone metric deltas. Negative values favor the fixed Student-t head. PatchTST NLL and CRPS improve on all three datasets, while Electricity has a small positive MSE delta.

optimization or gradient-flow failure.

The supported interpretation is narrower: the tested global-statistics FiLM conditioning adds no generalization value beyond matched static capacity in this fixed condition. It does not rule out local temporal or channel-structured conditioning. Detailed adapter results are retained in Appendix A.1.1.

7.2 Additive split conformal calibration

The conformal experiment uses a strict raw-time selection and calibration split with a 479-row embargo. Native test intervals are already near their nominal targets, while the calibration segment is descriptively harder. The evaluated additive correction has a nonnegative quantile and can only expand intervals. In this setting, expansion moves native test intervals toward overcoverage, increases width, and worsens both absolute coverage error and interval score.

Figure 6 displays the tradeoff. Mask-severity conditioning changes bin-specific quantiles but

Table 5: Controlled probability-head efficiency measurements. Timing comparisons are within each dataset and benchmark environment.

Dataset	Head	Params	Forward ms	NLL ms	CRPS ms	Peak MiB
etth1	Gaussian	6,168	29.997997	51.514500	86.934799	215.033691
etth1	Laplace	6,168	29.996887	30.016349	63.189127	215.033691
etth1	Fixed Student-t (df=8)	6,168	29.902914	65.900001	90.256716	215.034180
etth1	Learned Student-t	6,175	29.982518	69.371223	103.335362	215.034180
weather	Gaussian	24,624	59.357760	79.919755	188.685383	630.664551
weather	Laplace	24,624	58.485584	59.293318	133.341993	630.664551
weather	Fixed Student-t (df=8)	24,624	58.181018	98.037041	211.123562	630.665039
weather	Learned Student-t	24,645	59.026097	97.896011	212.516092	630.665039
electricity	Gaussian	24,624	356.894562	359.007118	1248.479182	4626.902832
electricity	Laplace	24,624	345.290164	343.346301	895.849843	4626.902832
electricity	Fixed Student-t (df=8)	24,624	343.877731	374.742924	1477.667372	4626.904297
electricity	Learned Student-t	24,945	420.384170	473.496327	1890.070674	4626.904297

Table 6: Within-backbone deltas for fixed Student-t versus Gaussian. The point threshold permits at most a 1% relative MSE increase.

Dataset	Backbone	ΔNLL	ΔCRPS	ΔMSE	Rel. MSE	NLL win	CRPS NI	Point NI	Full pair
ETTh1	TimePerceiver	-0.053587	-0.007578	-0.006405	-1.524%	Yes	Yes	Yes	Yes
ETTh1	PatchTST	-0.035317	-0.002538	-0.005439	-1.274%	Yes	Yes	Yes	Yes
Weather	TimePerceiver	-0.450160	-0.036981	-0.001243	-0.586%	Yes	Yes	Yes	Yes
Weather	PatchTST	-1.818698	-0.076205	-0.050179	-21.595%	Yes	Yes	Yes	Yes
Electricity	TimePerceiver	-0.073589	-0.001656	0.000592	+0.353%	Yes	Yes	Yes	Yes
Electricity	PatchTST	-0.083433	-0.001688	0.001980	+1.194%	Yes	Yes	No	No

does not repair the aggregate result. A test-label shrinkage calculation is retained only as a counterfactual diagnosis and is not a valid calibrator. This finding is specific to the additive score, split, dataset, and missingness condition.

7.3 PatchTST point-prediction tradeoff

The PatchTST Electricity result is not removed because its probabilistic scores improve. It is instead reported as a partial transfer result: NLL and CRPS improve, while relative MSE increases by 1.194%. This single threshold miss changes the complete second-backbone decision from positive to a point-prediction tradeoff.

8 Limitations

1. The study covers three datasets: ETTh1, Weather, and Electricity.
2. It evaluates two backbones, TimePerceiver and PatchTST.
3. Fixed test masks support pairing but do not represent every possible missingness realization.
4. Several expanded conditions use one probability-head training seed.

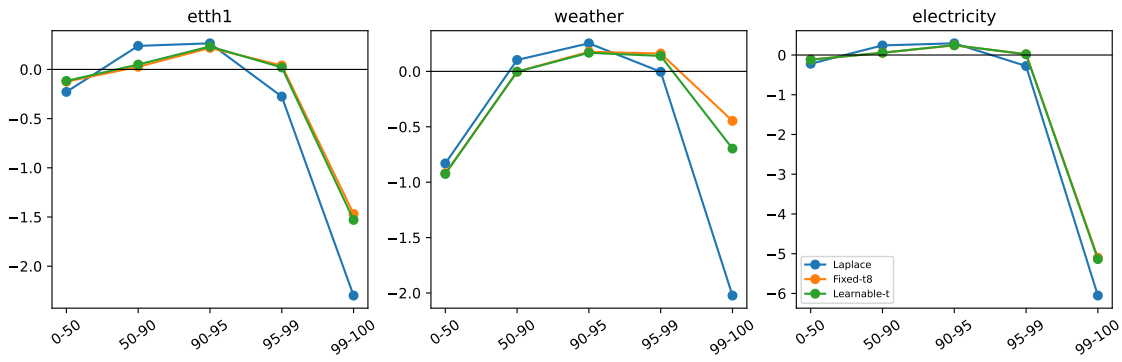


Figure 5: Descriptive NLL differences across Gaussian-residual percentile bins for the TimePerceiver distribution ablation.

5. The TimePerceiver multi-seed matrix keeps each official backbone checkpoint fixed.
6. PatchTST uses one clean-backbone seed and one probability-head seed.
7. The PatchTST study is limited to random missingness at rate 0.3.
8. No inferential significance analysis is reported.
9. The probability heads model marginal target-element distributions.
10. Cross-channel and cross-horizon covariance are not modeled.
11. Marginal interval coverage does not establish simultaneous channel or horizon coverage.
12. Fixed df=8 is empirically sufficient here but is not a theoretical optimum.
13. The complete second-backbone gate fails because Electricity MSE increases by 1.194%.
14. The conformal result concerns one additive split procedure on ETTh1 random-0.3.
15. The FiLM result concerns one global-statistics conditioning design and one insertion pattern.

These boundaries define the evidence rather than post-hoc exceptions. Broader backbone seeds, missingness realizations, distribution shifts, and covariance-aware heads remain open evaluation directions.

9 Conclusion

The evidence supports fixed heavy-tailed probabilistic forecasting under missing historical observations. The fixed missing-history protocol controls masks, normalization, model selection, and target access. On TimePerceiver, Student-t variants improve NLL and CRPS across three datasets and four probability-head training seeds. The distribution ablation supports fixed df=8 as sufficient under the evaluated setting.

On PatchTST, the fixed Student-t head improves NLL and CRPS on all three datasets. However, the Electricity point-prediction tradeoff prevents the complete second-backbone gate from being positive. The final evidence therefore supports probabilistic-score robustness across the two evaluated backbones, but it does not establish metric-wise invariance beyond them.

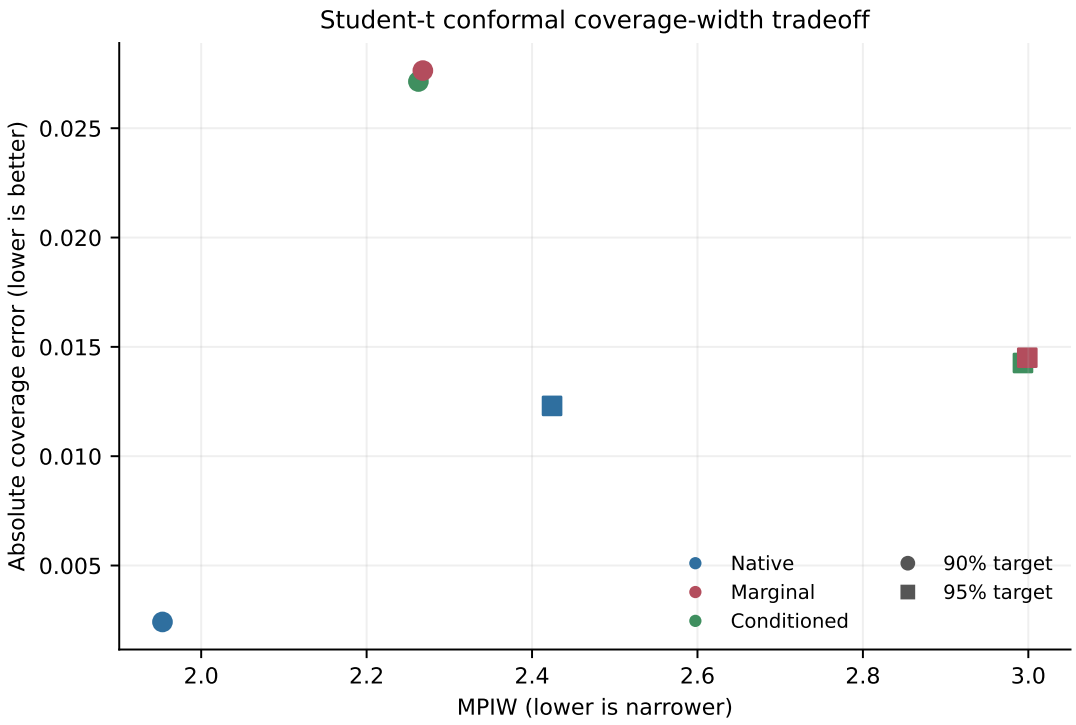


Figure 6: Native and additive conformal interval tradeoffs under the strict ETTh1 random-0.3 split. The additive procedures widen intervals without improving absolute coverage error.

References

[1] Wei Cao, Dong Wang, Jian Li, Hao Zhou, Lei Li, and Yitan Li. BRITS: Bidirectional recurrent imputation for time series. In *Advances in Neural Information Processing Systems*, volume 31, pages 6776–6786, 2018.

[2] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[3] Jaebin Lee and Hankook Lee. Timeperceiver: An encoder-decoder framework for generalized time-series forecasting. In *Advances in Neural Information Processing Systems*, volume 38, 2025.

[4] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations*, 2023.

[5] David Salinas, Valentin Flunkert, Jan Gasthaus, and Tim Januschowski. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3):1181–1191, 2020.

[6] Kamile Stankeviciute, Ahmed M. Alaa, and Mihaela van der Schaar. Conformal time-series forecasting. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[7] Yusuke Tashiro, Jiaming Song, Yang Song, and Stefano Ermon. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. In *Advances in Neural Information Processing Systems*, volume 34, pages 24804–24816, 2021.

A Additional Evidence and Reproducibility

A.1 Additional Results

A.1.1 Adapter ablation

Table 7 gives the full Gate 4a comparison. Figure 7 visualizes the same test MSE values. The missingness-aware FiLM branch is active, but it does not outperform the static adapter.

Table 7: Gate 4a adapter comparison on ETTh1 random-0.3.

Variant	Trainable	Overhead	Best epoch	Val. MSE	Test MSE	Test MAE
Official	0	0.00%	NA	NA	0.425394	0.450336
Static adapter	33,600	2.96%	1	0.678666	0.361377	0.390458
Missingness-aware FiLM adapter	39,136	3.44%	5	0.679754	0.363156	0.391077

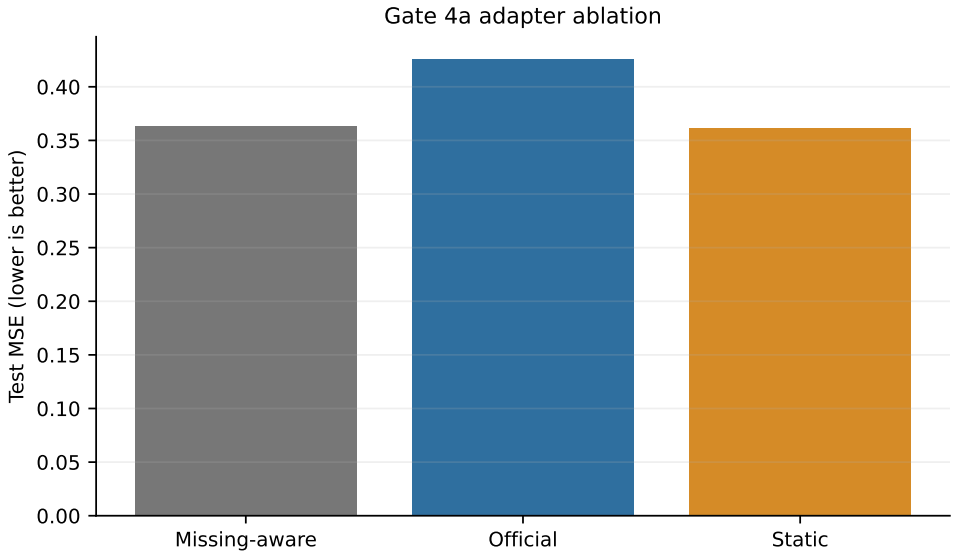


Figure 7: Static and missingness-aware adapter results relative to the official reference.

A.1.2 Cross-condition TimePerceiver evidence

Table 8 reports the ETTh1 condition matrix. It uses the same fixed evaluation assets for Gaussian and Student-t heads within each condition.

Table 8: ETTh1 cross-condition paired comparison.

Condition	ΔNLL	ΔCRPS	ΔMSE	NLL win	CRPS NI	Point NI
Random 0.1	-0.078025	-0.006113	-0.002231	Yes	Yes	Yes
Random 0.3	-0.041381	-0.003864	-0.001155	Yes	Yes	Yes
Random 0.5	-0.013720	0.000656	0.007100	Yes	No	No
Block 0.3	-0.061086	-0.007738	-0.004834	Yes	Yes	Yes
Channel 0.3	-0.071504	-0.004849	0.001022	Yes	Yes	Yes
Mixed 0.3	-0.070665	-0.003286	0.001169	Yes	Yes	Yes

A.1.3 Conformal details

Tables 9 and 10 retain the formal conformal results and mechanism audit. Figure 8 compares calibration and test coverage. Figures 9 and 9 show channel- and horizon-level diagnostics.

Table 9: Formal additive conformal results.

Method	Nominal	PICP	MPIW	Coverage error	Interval score
Mask-severity-conditioned conformal	0.90	0.927136	2.262613	0.027136	2.989775
Marginal conformal	0.90	0.927636	2.267989	0.027636	2.990528
Native	0.90	0.897581	1.953189	0.002419	2.946898
Mask-severity-conditioned conformal	0.95	0.964262	2.993500	0.014262	3.625242
Marginal conformal	0.95	0.964505	2.998730	0.014505	3.625259
Native	0.95	0.937700	2.424187	0.012300	3.596873

Table 10: Conformal mechanism audit. Counterfactual values are diagnostic and not formal calibration.

Head	Nominal	Cal. PICP	Test PICP	Shift	$\hat{q}$	Δ coverage error
Gaussian	0.90	0.869114	0.904533	0.035418	0.126842	0.023728
Gaussian	0.95	0.912017	0.937263	0.025246	0.279881	0.001974
Student-t	0.90	0.861667	0.897581	0.035913	0.157400	0.025216
Student-t	0.95	0.913178	0.937700	0.024522	0.287271	0.002205

A.1.4 Complete fixed-mask degradation matrix

Tables 11–13 retain every Gate 3b cell while splitting the matrix by dataset for readability.

A.1.5 All conformal heads

Table 14 reports every retained head and interval method. Extra probability diagnostics are shown in Figures 10–11.

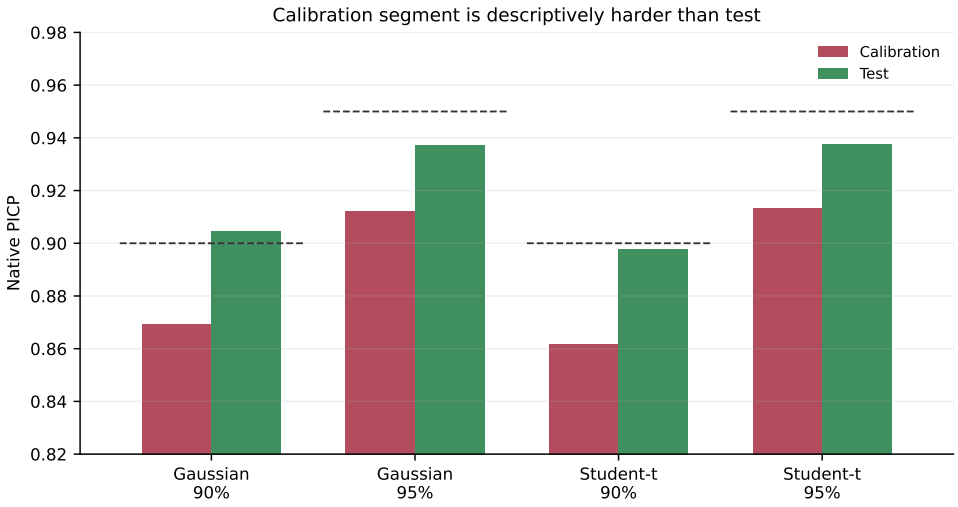


Figure 8: Descriptive calibration-to-test coverage shift under the strict temporal split.

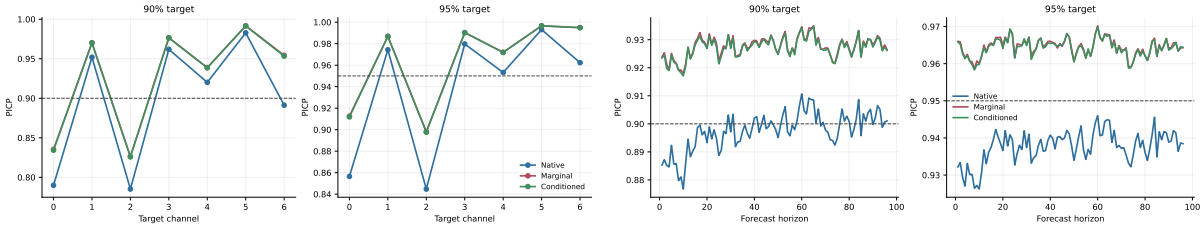


Figure 9: Channel-level and horizon-level coverage diagnostics for the Gate 4c experiment.

A.2 Protocol Details

The input length is 384 and the prediction length is 96 throughout the formal study. Dataset boundaries follow the official backbone implementations. Training-only normalization is frozen before validation or test access. Historical masks are applied after constructing the input window and before the backbone forward pass. Target windows are copied before masking and verified unchanged.

Random masks remove individual historical elements. Block masks remove contiguous temporal regions. Channel masks remove complete historical channels. Mixed masks combine mechanisms under the recorded rate. The test-mask seed is 2025. Matched methods load the same mask files and hashes.

Probability heads are selected by validation NLL. CRPS uses 64 MC samples and seed 52025. Native intervals use each distribution's marginal quantiles. The Gate 4c calibration segment follows the selection segment and is separated by a 479-row embargo. Test labels are excluded from calibration.

The paired point criterion allows at most a 1% relative MSE increase. This criterion is applied identically to TimePerceiver and PatchTST. It is the reason PatchTST Electricity remains a failed point-noninferiority cell despite improved NLL and CRPS.

Table 11: Full ETTh1 fixed-mask degradation matrix.

Mask	Rate	MSE	MAE	RMSE
Block	0.10	0.388818	0.416915	0.623552
Block	0.30	0.423837	0.451188	0.651027
Block	0.50	0.486504	0.501242	0.697498
Block	0.70	0.591697	0.571655	0.769219
Channel	0.10	0.479410	0.455063	0.692394
Channel	0.30	0.583704	0.511084	0.764006
Channel	0.50	0.798862	0.627325	0.893791
Channel	0.70	0.900947	0.683061	0.949182
Mixed	0.10	0.486209	0.464128	0.697287
Mixed	0.30	0.495055	0.473908	0.703601
Mixed	0.50	0.553485	0.525226	0.743966
Mixed	0.70	0.715806	0.617339	0.846053
Random	0.10	0.378811	0.408069	0.615476
Random	0.30	0.425394	0.450336	0.652223
Random	0.50	0.525856	0.523115	0.725160
Random	0.70	0.688520	0.617690	0.829771

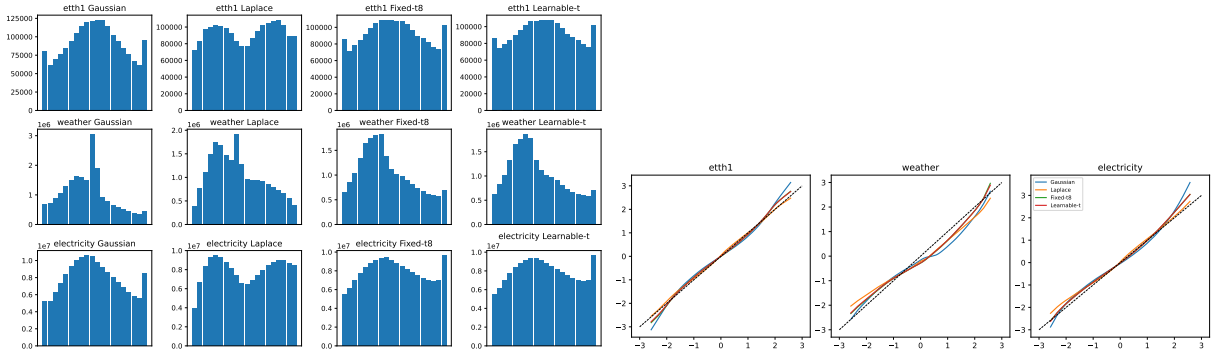


Figure 10: Descriptive probability integral transform and quantile diagnostics.

A.3 Reproducibility Record

The anonymous evidence package records the resolved configuration, random seed, checkpoint hash, validation-mask hash, test-mask hash, training-protocol hash, metric-protocol hash, and source row for every formal paper result. Invalid historical runs that did not apply the input mask remain in the internal audit registry and are excluded from all manuscript tables, figures, claims, and reproduction assets.

The TimePerceiver source commit is `7e30cc07b51c709f408409fd60a34c81ae8990be`. The PatchTST source commit is `204c21efe0b39603ad6e2ca640ef5896646ab1a9`. The recorded software environment uses Python 3.9.25, PyTorch 2.1.0 with CUDA runtime 12.1, and fixed metric implementations. Public code locations will replace the anonymous placeholder after review.

Table 12: Full Weather fixed-mask degradation matrix.

Mask	Rate	MSE	MAE	RMSE
Block	0.10	0.158302	0.218956	0.397872
Block	0.30	0.180420	0.245178	0.424759
Block	0.50	0.210474	0.277453	0.458774
Block	0.70	0.254572	0.322297	0.504551
Channel	0.10	0.191289	0.238851	0.437366
Channel	0.30	0.284471	0.316355	0.533358
Channel	0.50	0.377902	0.393866	0.614738
Channel	0.70	0.495709	0.490618	0.704066
Mixed	0.10	0.175815	0.230739	0.419303
Mixed	0.30	0.224939	0.281431	0.474278
Mixed	0.50	0.306930	0.357264	0.554012
Mixed	0.70	0.400193	0.440502	0.632608
Random	0.10	0.169631	0.232543	0.411863
Random	0.30	0.240810	0.309629	0.490724
Random	0.50	0.331313	0.391138	0.575598
Random	0.70	0.434543	0.474367	0.659199

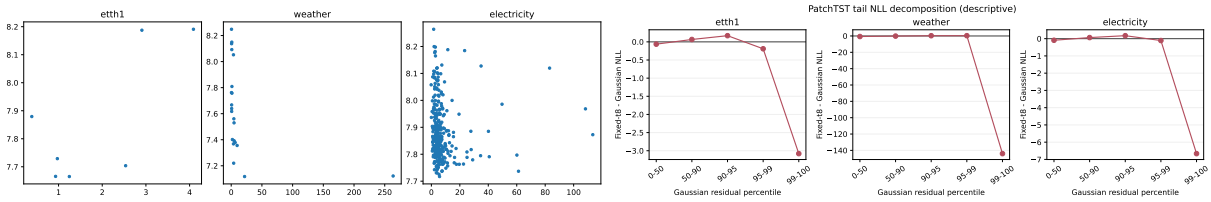


Figure 11: Channel-tail diagnostics and the descriptive PatchTST residual-percentile NLL decomposition.

Table 13: Full Electricity fixed-mask degradation matrix.

Mask	Rate	MSE	MAE	RMSE
Block	0.10	0.143247	0.243795	0.378480
Block	0.30	0.184630	0.289155	0.429686
Block	0.50	0.270835	0.369677	0.520418
Block	0.70	0.464169	0.520260	0.681300
Channel	0.10	0.213972	0.281108	0.462571
Channel	0.30	0.391038	0.403166	0.625331
Channel	0.50	0.569116	0.525639	0.754398
Channel	0.70	0.748916	0.649662	0.865399
Mixed	0.10	0.165759	0.254616	0.407135
Mixed	0.30	0.246785	0.325159	0.496775
Mixed	0.50	0.375182	0.437690	0.612521
Mixed	0.70	0.588662	0.596810	0.767243
Random	0.10	0.138429	0.242728	0.372060
Random	0.30	0.200572	0.318891	0.447853
Random	0.50	0.325732	0.432183	0.570729
Random	0.70	0.546080	0.584678	0.738972

Table 14: All Gate 4c head and interval-method combinations.

Head	Method	Nominal	PICP	MPIW	Coverage error	Interval score
Gaussian	Mask-severity-conditioned conformal	0.90	0.927797	2.289391	0.027797	2.979145
Gaussian	Marginal conformal	0.90	0.928261	2.294196	0.028261	2.979930
Gaussian	Native	0.90	0.904533	2.040512	0.004533	2.936425
Gaussian	Mask-severity-conditioned conformal	0.95	0.964530	2.986506	0.014530	3.598434
Gaussian	Marginal conformal	0.95	0.964711	2.991181	0.014711	3.598406
Gaussian	Native	0.95	0.937263	2.431420	0.012737	3.571486
Student-t	Mask-severity-conditioned conformal	0.90	0.927136	2.262613	0.027136	2.989775
Student-t	Marginal conformal	0.90	0.927636	2.267989	0.027636	2.990528
Student-t	Native	0.90	0.897581	1.953189	0.002419	2.946898
Student-t	Mask-severity-conditioned conformal	0.95	0.964262	2.993500	0.014262	3.625242
Student-t	Marginal conformal	0.95	0.964505	2.998730	0.014505	3.625259
Student-t	Native	0.95	0.937700	2.424187	0.012300	3.596873